Business Case & Policy Consulting Theory Handbook

Cost-benefit analysis, evaluation, market analysis and interview preparation
Sector module: the EU space ecosystem

How to use this handbook

- **Part 1–2** give you the domain: who does what in EU space, how the systems work, which policies matter.
- **Part 3–6** give you the methods the team uses every day: cost-benefit analysis, evaluation, market analysis, uptake.
- **Part 7–9** give you tools and the interview playbook.
- Each chapter ends with **"How to say it"**: a short sentence you can use in the interview.
- Facts were checked in November 2026 against public sources (European Commission, EUSPA, ESA, European Court of Auditors). Items marked **[check]** change often: verify them the day before the interview.

Written for personal study. All worked examples use illustrative numbers unless a source is named.

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1. The role and the team

The reference role is a **junior consultant** in a market-intelligence and policy team of a consultancy working for EU institutions. Such a team advises European institutions and private clients on space and telecommunications, digital and transport. In space, its typical clients are **EUSPA**, **DG DEFIS**, **ESA** and other EU bodies active in upstream and downstream space.

What the team sells

Service	What it means in practice
Market analysis	Size and forecast markets for satellite navigation and Earth Observation services, by segment (agriculture, road, aviation…).
User needs and requirements	Find out what users need (accuracy, timeliness, price, format) and translate it into requirements for EU systems.
Cost-benefit and economic analysis	Decide if an EU space initiative is worth its cost for society.
Feasibility assessments	Check whether a new space initiative is technically, economically and institutionally viable.
Policy support	Evaluations, impact assessments, strategies, KPIs and monitoring frameworks.

What a junior actually does

- **Desk research**: collect data from reports, company accounts, databases, regulation.
- **Primary research**: prepare and run interviews and surveys, write up case studies.
- **Modelling**: build and update Excel models (market forecasts, CBA).
- **Drafting**: sections of reports, slides, meeting notes.
- **Quality**: check figures, sources, consistency between chapters.

In the space sector module, the focus is on **secure satcom, Copernicus and Galileo data and services**, with quantitative *and* qualitative assessments.

> **What they test in the interview**
>
> 1. Can you **structure a quantitative problem** (CBA, market estimate) out loud?
> 2. Do you **know the domain** well enough to talk to EUSPA without embarrassment?
> 3. Have you **managed a piece of research** from start to finish?
> 4. Do you use **AI** in a concrete and critical way?
> 5. Can you **communicate clearly in English**?

2. The EU space ecosystem

2.1 Institutions: who does what

Body	Role
European Commission – DG DEFIS Directorate-General for Defence Industry and Space, created in 2020	Owner of the EU Space Programme. Sets policy, proposes legislation and the budget, manages the programme overall, signs the big contracts (e.g. the IRIS² concession). Political lead: the Commissioner for Defence and Space.
EUSPA EU Agency for the Space Programme, Prague; successor of the GSA (European GNSS Agency)	Operational exploitation of Galileo and EGNOS; security accreditation of all components; **market development and user uptake** for Galileo, EGNOS and Copernicus; roles in GOVSATCOM and IRIS². Publishes the *EU Space Market Report* and User Requirements Reports. **The team's main client.**
ESA European Space Agency, Paris; intergovernmental	**Not an EU body**: its members include non-EU countries (e.g. UK, Switzerland, Norway). For the EU programme it is design authority and procurement agent for many space components (Galileo satellites, Copernicus Sentinels). Runs its own programmes: science, launchers, technology, navigation demonstrators such as **Celeste** (LEO-PNT).
EU SatCen Torrejón, Spain	Geospatial intelligence for EU security and defence; involved in the Copernicus security service and in space surveillance.
Copernicus entrusted entities	Organisations that run the six Copernicus services: ECMWF (atmosphere – CAMS; climate change – C3S), Mercator Ocean International (marine – CMEMS), EEA and JRC (land; emergency management – CEMS), Frontex, EMSA and SatCen (security).
European Court of Auditors	Audits EU spending. Special Report 07/2021 on Galileo and Copernicus uptake is directly relevant to the team's work.
Member States and national agencies	CNES (France), DLR (Germany), ASI (Italy) and others: national programmes, ESA contributions, national users of EU services.
Industry	Large primes (e.g. Airbus, Thales Alenia Space, OHB), satellite operators (SES, Eutelsat, Hispasat), launch providers (Arianespace, Avio), and many New Space companies.

> *"DG DEFIS owns the programme and the policy, EUSPA operates Galileo and EGNOS and drives market uptake, and ESA is the intergovernmental technical partner that designs and procures much of the space infrastructure."*

2.2 The EU Space Programme 2021–2027

Established by **Regulation (EU) 2021/696**, which brought all EU space activities under a single programme and turned the GSA into EUSPA.

Budget 2021–2027 (current prices)

Component	Budget	What it is

Galileo and EGNOS	≈ €9.0bn	Satellite navigation and its European augmentation
Copernicus	≈ €5.4bn	Earth Observation
SSA and GOVSATCOM	≈ €0.44bn	Space situational awareness; governmental satellite communications
Total	**€14.88bn**	

IRIS² (secure connectivity) has its own regulation, **(EU) 2023/588**, and its own financing.

> **Next budget, 2028–2034 [check]**
> The Commission's proposal of July 2025 puts space inside a new **European Competitiveness Fund**, in a window called "Resilience, Security, Defence Industry and Space" of about €131bn. Space has **no ring-fenced amount**; industry associations have asked for one. The proposal needs unanimity in the Council and the consent of the Parliament, so the final figures may differ.

2.3 Satellite navigation (GNSS)

How it works

Each navigation satellite broadcasts a signal with its precise position and the time from an atomic clock. A receiver measures how long each signal took to arrive and converts it into a distance (a "pseudorange"). With distances to several satellites it computes its position by **trilateration**.

- You need **at least four satellites**: three unknowns for position (x, y, z) plus one for the error of the receiver's cheap clock.
- Positioning, Navigation and **Timing (PNT)**: timing is just as important as position. Telecom networks, power grids and financial markets synchronise their clocks with GNSS.
- Main **error sources**: ionosphere and troposphere delays, satellite clock and orbit errors, multipath (reflections off buildings), poor geometry, interference.

Improving accuracy and trust

Technique	How it works	Typical accuracy
Standalone, single frequency	Basic smartphone positioning	a few metres
Dual frequency	Removes most of the ionospheric error (Galileo E1 + E5)	better than standalone, under a few metres
SBAS (e.g. EGNOS)	Geostationary satellites broadcast corrections and **integrity** information over a region	1–2 m, with guaranteed integrity for aviation
PPP (e.g. Galileo HAS)	Precise orbit and clock corrections from satellites; needs time to converge	decimetres (≈20 cm for HAS)
RTK	Corrections from a nearby reference station	centimetres

Integrity is the ability to warn the user when the position should not be trusted. It matters for safety-of-life uses such as aviation.

Threats: jamming and spoofing

- **Jamming**: transmitting noise so the receiver loses the very weak GNSS signal. Widespread interference has affected aviation in the Baltic region and the Middle East.
- **Spoofing**: transmitting fake signals so the receiver computes a wrong position or time without noticing. More dangerous because it is silent.
- Countermeasures: authentication (Galileo **OSNMA**), encrypted signals (**PRS**), multi-constellation receivers, antennas that reject interference, alternative PNT (LEO-PNT, terrestrial systems, inertial sensors).

Galileo

The EU's global navigation system, under **civilian control**. Satellites in medium Earth orbit (about 23,200 km). A second generation (G2) is in development.

Service	Description	Status
Open Service (OS)	Free positioning and timing for everyone; in billions of smartphones and in cars.	Operational
High Accuracy Service (HAS)	Free precise corrections (PPP), around 20 cm.	Initial service since January 2023
OSNMA	Open Service Navigation Message Authentication: lets receivers check that the navigation data really come from Galileo. Defence against spoofing.	Initial service declared 24 July 2025
Public Regulated Service (PRS)	Encrypted, robust service for government-authorised users (civil protection, police, defence).	Restricted users
Search and Rescue (SAR)	Detects distress beacons; Galileo adds a **return link** telling the person in distress that the alert was received.	Operational

EGNOS

European SBAS. It monitors GPS (and in future Galileo, with EGNOS V3) and broadcasts corrections and integrity over Europe through geostationary satellites. Main use: **aircraft approaches** (LPV procedures) without ground equipment at the airport; also agriculture and maritime.

Why Galileo if GPS is free?

1. **Strategic autonomy**: GPS belongs to the US military; access could be degraded in a crisis.
2. **Performance**: with GPS + Galileo (+ others) receivers see more satellites, so accuracy and availability improve, especially in "urban canyons".
3. **Distinctive services**: HAS, OSNMA, SAR return link, PRS.
4. **Industrial and economic**: European industry and jobs, and a seat at the table in international standards.

> **The honest economic point**
> Most receivers are multi-constellation. The economic value of Galileo is therefore its **marginal** value over GPS and the others, not the value of positioning as a whole. Raise this yourself: it shows you understand CBA (see section 3.6).

LEO-PNT

Positioning, navigation and timing from **low Earth orbit** (roughly 500–1,200 km) instead of medium orbit.

- **Advantages**: signals arrive much stronger (shorter distance), so they resist jamming better; satellites move fast across the sky, so precise solutions converge faster; possible new frequencies.
- **Disadvantages**: many more satellites are needed for global coverage; shorter satellite life; very precise orbit and clock determination is needed.
- **Role**: a **complement** to Galileo for resilience and performance, not a replacement.
- **ESA Celeste**: in-orbit demonstrator. The first two satellites were launched at the end of March 2026 on a Rocket Lab Electron; about eight more are planned from 2027.
- Commercial players are also developing LEO-PNT services. In any EU CBA they belong in the options or the counterfactual.

> *"Galileo gives Europe autonomy and services GPS doesn't offer, like authentication and free high accuracy. LEO-PNT would add resilience against jamming and spoofing, which is now a real operational problem, but the CBA has to compare it with what Galileo and commercial providers already deliver."*

2.4 Earth Observation (Copernicus)

Basics of Earth Observation

Concept	Meaning
Passive optical	Measures sunlight reflected by the surface in several bands (visible, near-infrared, short-wave infrared). Blocked by clouds and darkness.
Active radar (SAR)	Sends microwaves and measures the echo. Works **day and night and through clouds**. Good for floods, ships, ground deformation (interferometry, InSAR), sea ice.
Spatial resolution	Size of a pixel on the ground (e.g. 10 m for Sentinel-2; below 1 m for commercial very-high-resolution satellites).
Spectral resolution	Number and width of bands. More bands allow more indices: vegetation, water, minerals.
Temporal resolution (revisit)	How often the same place is observed. Crucial for agriculture, disasters, monitoring of activity.
Sun-synchronous orbit	Low polar orbit that passes over each place at the same local time: consistent lighting for comparisons.
NDVI	Vegetation index = (NIR − Red) / (NIR + Red). Healthy vegetation reflects strongly in near-infrared.

Copernicus in one page

The EU's Earth Observation programme, with a **free, full and open data policy**: anyone can use the data for any purpose. Access through the **Copernicus Data Space Ecosystem**.

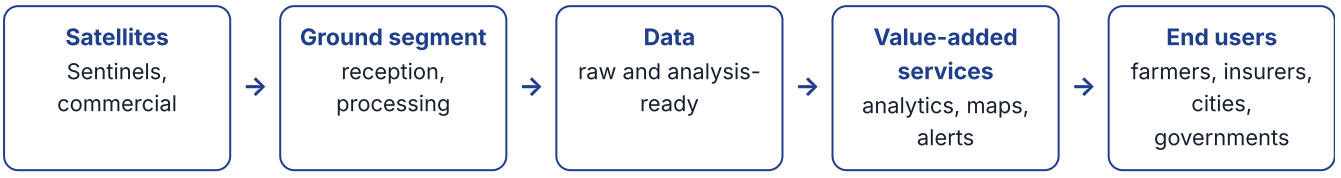
Mission	Instrument	Main uses
Sentinel-1	C-band SAR radar	Floods, ships and oil spills, ice, ground motion, all-weather monitoring

Sentinel-2	Multispectral optical, 13 bands, 10–60 m	Agriculture, forests, land use, water quality
Sentinel-3	Ocean and land colour, surface temperature, altimetry	Oceans, climate, wildfires, inland water
Sentinel-5P (and 4, 5)	Atmospheric spectrometers	Air quality, greenhouse and trace gases
Sentinel-6	Radar altimeter	Global sea level
Expansion missions	CO2M, CHIME, CIMR, CRISTAL, LSTM, ROSE-L	CO_2 emissions, hyperspectral, polar, thermal, L-band radar

The six Copernicus services

Service	Run by	Examples
Atmosphere (CAMS)	ECMWF	Air quality forecasts, UV, greenhouse gases
Marine (CMEMS)	Mercator Ocean International	Ocean temperature, currents, sea ice
Land (CLMS)	EEA, JRC	Land cover, vegetation, water bodies
Climate Change (C3S)	ECMWF	Climate reanalysis, seasonal forecasts
Emergency Management (CEMS)	JRC	Rapid mapping of floods, fires, earthquakes; early warning
Security	Frontex, EMSA, SatCen	Border and maritime surveillance, support to EU external action

The EO value chain



Most of the economic value is created at the right end of the chain. Free data are necessary but not sufficient: users need processed, ready-to-use information.

Sentinel vs commercial very-high-resolution data

Copernicus also buys data from **contributing missions**, often commercial satellites with sub-metre resolution. New Space constellations now offer very high resolution with daily or even sub-daily revisit.

When is paying worth it? Only when the extra resolution or revisit **changes a decision**: small objects (vehicles, buildings), urban change, infrastructure inspection, insurance claims, emergencies where hours matter. For crop monitoring at field level, Sentinel-2 at 10 m is often enough. This is a **value-of-information** question (section 3.5).

> *"Copernicus data are free and open, but uptake depends on turning data into ready-to-use services. Commercial high-resolution data add value only where they change a decision, so I would assess them with a value-of-information approach."*

2.5 Secure satellite communications

Orbits and trade-offs

Orbit	Altitude	Coverage per satellite	Latency	Satellites needed
GEO	≈ 35,786 km	very large, fixed over one area	high (≈ 600 ms round trip)	few
MEO	≈ 8,000–20,000 km	large	medium	tens
LEO	≈ 500–1,200 km	small, moving	low (tens of ms)	hundreds to thousands

GOVSATCOM

Pooling and sharing of **existing** secure satellite capacity (national military and commercial) for EU and Member State governmental users: crisis management, surveillance, key infrastructure. A component of the 2021–2027 programme, with ground "hubs" connecting users to capacity.

IRIS²

Infrastructure for Resilience, Interconnectivity and Security by Satellite. An EU-owned **multi-orbit constellation** (LEO and MEO) for secure governmental communications, with commercial services on the same infrastructure.

- **Delivery model**: 12-year **concession** to the SpaceRISE consortium (SES, Eutelsat, Hispasat), signed on 16 December 2024. A public-private partnership: the EU and ESA finance part, the private partners finance and operate the rest and sell commercial services.
- **Initial plan**: about 290 satellites, about €10.6bn in total, of which about €6.5bn public; governmental services targeted for 2030.
- **[check]** In 2026 the EU and SpaceRISE announced a reinforcement (more satellites, more governmental capacity) and an acceleration to 2029. Press figures on satellites and budget vary.
- **Rationale**: sovereignty, security of communications, reducing dependence on non-EU systems such as Starlink (highlighted by the war in Ukraine), connectivity in remote areas, quantum-encryption readiness.

EUSPA's 2026 market report estimates EU demand for secure satcom data services growing from just over €0.2bn (2025) to about €1.2bn (2040), led today by maritime surveillance and later by law enforcement, civil protection and force deployment.

2.6 Space situational awareness (SSA)

- **Space Surveillance and Tracking (EU SST)**: a partnership of Member States providing collision-avoidance warnings, re-entry and fragmentation analysis.
- **Space weather**: solar activity that can disrupt GNSS, satellites and power grids.
- **Near-Earth objects**: asteroids and comets.

The issue is growing fast with mega-constellations and debris: the EU Space Act (below) addresses it through regulation.

2.7 Policies and documents to know

EU Space Act (proposal)

- Commission proposal **COM(2025) 335**, 25 June 2025: a regulation on the safety, resilience and sustainability of space activities.
- **Safety**: tracking, collision avoidance, debris mitigation, end-of-life rules.
- **Resilience**: cybersecurity and risk management for space operators, as a *lex specialis* to the NIS2 Directive.
- **Sustainability**: a common method to measure the environmental footprint of space activities.
- Applies to EU and non-EU operators serving the EU market; expected to apply from 2030. **[check]** Status in Parliament (ITRE committee) and Council.
- For a consultancy: impact assessments, compliance-cost studies, support to implementation.

European Court of Auditors, Special Report 07/2021

"EU space programmes Galileo and Copernicus: services launched, but the uptake needs a further boost".

- Copernicus data used mainly by experts and scientists; Commission actions had not produced the hoped-for rise in uptake.
- Weak monitoring of Galileo uptake; some key features not yet available at the time.
- **No comprehensive strategy with measurable, time-bound objectives.**
- Recommendations: a comprehensive uptake strategy; a framework to assess economic and societal benefits; integrated access to Copernicus data; identification of regulatory and administrative barriers.
- Follow-up: in 2024 the Commission published a staff working document on uptake with a conceptual framework to measure it.

Supporting EUSPA in implementing these recommendations is a typical assignment: you should be able to propose **KPIs** (section 4.4).

Other references

- **EU Space Strategy for Security and Defence** (March 2023): space as a security domain, protection of space assets, resilience.
- **European Chips Act** (2023): relevant to the project on semiconductors for defence and space.
- **EUSPA EU Space Market Report** (May 2026): the reference for downstream market figures.

2.8 The space economy

Segment	Definition	Examples
Upstream	Designing, building and launching space infrastructure	Satellite manufacturing, launchers, ground stations
Midstream	Operating satellites and distributing data and signals (sometimes included in upstream or downstream)	Satellite operators, ground segment services, data platforms
Downstream	Products and services that use space signals and data	GNSS receivers and apps, EO analytics, satcom services

In-space economy	Activities that take place in orbit	Servicing and refuelling satellites, in-orbit manufacturing, debris removal, space logistics

New Space

Private, often venture-funded companies; small satellites in large constellations; lower launch costs; vertical integration; "data as a service" business models. Consequences for the EU: more commercial offers to buy instead of build; competition with large incumbents; questions of sovereignty when key providers are non-European. Launchers: Ariane 6 flew for the first time in July 2024; European launch autonomy remains a policy priority.

Market figures: EUSPA EU Space Market Report 2026

The 2026 edition (May 2026) brings together **GNSS, Earth Observation, secure satcom and SSA** in one report with **16 market segments**; the new segment is **EU border and internal security**.

Market	2024	Forecast	Key points
GNSS downstream (global revenues)	≈ €300bn	≈ €580bn (2034)	Led by consumer solutions and road/automotive. Installed base near 10bn devices by 2034. Services ≈ 80% of revenues by 2034.
Earth Observation (data and value-added services)	≈ €3.5bn	≈ €7.9bn (2034)	Agriculture largest segment (≈ 21% in 2024). European suppliers hold ≈ 42% of revenues.
Secure satcom (EU user data services)	≈ €0.2bn (2025)	≈ €1.2bn (2040)	Maritime surveillance today; law enforcement, civil protection, force deployment later.

Implied growth rates: GNSS $(580/300)^{1/10} - 1 \approx$ **6.8%** a year; EO $(7.9/3.5)^{1/10} - 1 \approx$ **8.5%** a year. Being able to compute this on the spot is a good signal.

Segments in EUSPA reports include: agriculture; aviation and drones; climate, environment and biodiversity; consumer solutions, tourism and health; emergency management and humanitarian aid; energy and raw materials; fisheries and aquaculture; forestry; infrastructure; insurance and finance; maritime and inland waterways; rail; road and automotive; urban development and cultural heritage; space; EU border and internal security.

> *"Downstream is where the money and the benefits are: GNSS alone is around €300 billion a year and EUSPA expects it to nearly double by 2034, mostly through services rather than devices."*

3. Cost-benefit analysis

Cost-benefit analysis (CBA) measures whether an intervention **increases the welfare of society**. It is not about whether the project is profitable for the operator; it is about whether society is better off with it than without it. Reference: the European Commission's *Guide to Cost-Benefit Analysis of Investment Projects* (2014) and the *Better Regulation Toolbox*.

3.1 Time value of money

€100 today is worth more than €100 in ten years: money can be invested, people prefer consumption now, and the future is uncertain. To compare flows at different dates we **discount** them to their **present value**.

$$PV = X / (1 + r)^t \qquad \text{Discount factor } DF_t = 1 / (1 + r)^t$$

$$\text{Annuity (constant } X \text{ for years } 1\ldots n\text{): } PV = X \times [1 - (1 + r)^{-n}] / r$$

Example: €100m received in year 10 at 3% is worth $100 / 1.03^{10} = 100 / 1.344 =$ **€74.4m** today. Discounting is **compound**, not linear: 100 × (1 − 0.03 × 10) = €70m is wrong.

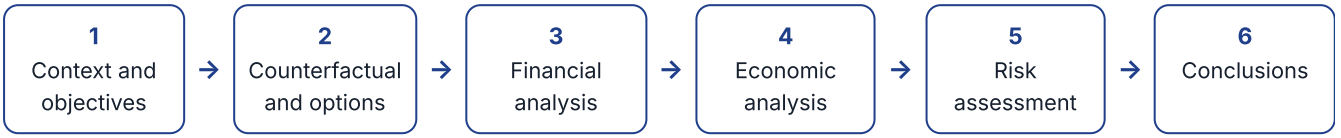
Real vs nominal

- **Constant prices** (inflation removed) → use a **real** discount rate.
- **Current prices** (including inflation) → use a **nominal** rate.
- Link: (1 + nominal) = (1 + real) × (1 + inflation). Never mix the two.

Which discount rate?

- **Social discount rate** (economic analysis): the 2014 Commission Guide uses a benchmark of **3% real**, and **5%** for cohesion countries; the Better Regulation Toolbox recommends 3%.
- **Financial discount rate**: benchmark of 4% real in the 2014 Guide.
- A higher rate penalises projects whose benefits come late. Space infrastructure has high upfront costs and long benefit streams, so results are sensitive to the rate.

3.2 The CBA method step by step



Step 1 – Context and objectives

What problem does the intervention address? What are the objectives, ideally quantified? Example: "reduce the economic exposure of critical infrastructure to GNSS disruption".

Step 2 – Counterfactual and options

The **counterfactual** (or baseline) is what happens **without** the intervention. Everything in a CBA is measured as the **difference** between the scenario with the project and the counterfactual: the **incremental approach**.

- **Do-nothing**: no new action; existing assets keep running.
- **Do-minimum**: only what is needed to keep the current service level.
- **Options**: realistic alternatives (e.g. build an EU system, buy a commercial service, upgrade existing systems). A CBA of only one option against do-nothing is weak.

> **Typical mistake**
> Choosing a counterfactual that is too pessimistic makes any project look good. In space, the counterfactual must include what Galileo, GPS and commercial providers will deliver anyway.

Step 3 – Financial analysis

From the point of view of the **owner or operator**: investment, operating costs, revenues, residual value. Indicators: financial net present value (**FNPV**) and financial rate of return (**FRR**). It shows if the project needs public funding and if it is financially sustainable (cash never runs out). Public space infrastructure usually has a **negative FNPV**: that is why it is public.

Step 4 – Economic analysis

From the point of view of **society**. Starting from the financial flows:

1. **Remove transfers**: taxes, subsidies and other payments between parts of society do not create or destroy resources.
2. **Apply shadow prices** (conversion factors) where market prices are distorted, e.g. labour in areas of high unemployment.
3. **Add externalities** and **non-market benefits**: time saved, lives saved, emissions avoided, damage avoided.
4. **Discount** at the social discount rate.

$$ENPV = \Sigma_t \, (B_t - C_t) \, / \, (1 + r)^t$$

$$B/C \text{ ratio} = PV(\text{benefits}) \, / \, PV(\text{costs})$$

$$ERR = \text{the discount rate at which } ENPV = 0$$

Indicator	Rule	Strengths and limits
ENPV	> 0	The main indicator: absolute net gain for society. Depends on project size.
B/C ratio	> 1	Easy to communicate; independent of size. Sensitive to whether a cost is netted from benefits or added to costs.
ERR (IRR)	> social discount rate	Intuitive. Can be misleading: multiple IRRs if flows change sign several times; does not rank projects of different size well.

Other elements

- **Time horizon**: the economic life of the asset (for satellites, typically 10–15 years per generation; for systems, longer with replenishment).
- **Residual value**: value of assets at the end of the horizon, if any.
- **Sunk costs**: costs already spent are **excluded**; only future, avoidable costs count.
- **Replacement and replenishment**: satellites must be replaced; forgetting this is a classic error.

3.3 Worked example

Illustrative case: a public flood-mapping service based on Earth Observation for a river basin. Investment €12m in year 0; operating costs €1.5m a year; avoided flood damage (benefit) €3.2m a year; years 1–10; social discount rate 3% real; no residual value.

Cash flows and discounting (€m)

Year	Costs	Benefits	Net flow	DF at 3%	Discounted net flow
0	12.0	0.0	-12.0	1.0000	-12.00
1	1.5	3.2	1.7	0.9709	1.65
2	1.5	3.2	1.7	0.9426	1.60
3	1.5	3.2	1.7	0.9151	1.56
4	1.5	3.2	1.7	0.8885	1.51
5	1.5	3.2	1.7	0.8626	1.47
6	1.5	3.2	1.7	0.8375	1.42
7	1.5	3.2	1.7	0.8131	1.38
8	1.5	3.2	1.7	0.7894	1.34
9	1.5	3.2	1.7	0.7664	1.30
10	1.5	3.2	1.7	0.7441	1.27
				ENPV	+2.50

- Annuity factor (3%, 10 years) = 8.530.
- PV of benefits = 3.2 × 8.530 = **€27.30m**. PV of costs = 12 + 1.5 × 8.530 = **€24.80m**.
- **ENPV = +€2.50m**; **B/C = 1.10**; **ERR ≈ 6.9%** (above 3%).

Conclusion at first sight: the project is worth doing. But look at the risk analysis.

3.4 Risk and sensitivity analysis

Sensitivity analysis

Change one variable at a time and see how the result moves. The 2014 Guide defines a variable as **critical** when a **1% change** in it causes a change of **more than 1%** in the NPV.

Sensitivity of the worked example

Scenario	ENPV (€m)	B/C
Base case	+2.50	1.10
Benefits −10%	-0.23	0.99
Benefits −20%	-2.96	0.88
Operating costs +20%	-0.06	1.00

Investment +20%	+0.10	1.00
Discount rate 5% instead of 3%	+1.13	1.05

All variables are critical here: a 1% fall in benefits reduces ENPV by €0.27m, about 11% of €2.50m.

Switching values

The **switching value** is the percentage change in a variable that brings the NPV to zero.

```
Switching value of benefits = ENPV / PV(benefits) = 2.50 / 27.30 = 9.2%
Switching value of operating costs = ENPV / PV(opex) = 2.50 / 12.80 = 19.5%
Switching value of investment = ENPV / investment = 2.50 / 12.0 = 20.8%
```

Interpretation: if avoided damage is overestimated by just 9.2%, the project no longer pays off. Given how uncertain flood damage estimates are, the right message to the client is "positive but fragile; the result depends on the benefit estimate, which needs better evidence".

Scenario and probabilistic analysis

- **Scenario analysis**: change several variables together in consistent stories (pessimistic, base, optimistic).
- **Monte Carlo**: give each critical variable a probability distribution, draw thousands of combinations, and report the distribution of NPV, e.g. "probability that ENPV < 0 is 35%".
- **Qualitative risk assessment**: risk matrix (probability × impact) with mitigation measures, e.g. launch failure, delays, low user adoption.

> *"I'd always report switching values, because a positive NPV that turns negative with a 9% error in benefits is a very different message from one that survives a 50% error."*

3.5 Valuing benefits

Benefit categories in space projects

Category	Description	Example
Direct (industry)	Revenues, jobs and value added of companies building and operating the system	Satellite manufacturing contracts
Indirect and induced	Supply chain effects; spending of the income generated	Suppliers of components; household spending
User benefits	Savings and gains for those who use the services	Fuel saved in precision agriculture; time saved in logistics
Wider societal	Benefits to society beyond direct users	Lives saved by search and rescue; lower emissions
Strategic	Autonomy, security, resilience	Continuity of timing for power grids during GNSS disruption

> **Careful with industry effects**
> In a **CBA**, jobs and revenues of the space industry are mostly **not benefits**: the money spent on the project is a cost, and the workers would otherwise be employed elsewhere. They belong in an **economic impact study** (section 3.8), not in the ENPV. Mixing the two is a common error.

Valuation methods

Method	Idea	Space example
Avoided cost / damage	Value = losses that do not happen thanks to the service	Flood damage avoided through early mapping; outage losses avoided by resilient timing
Productivity / cost savings	Value = lower inputs or higher outputs	Fertiliser and fuel saved with precision farming; higher yields
Time savings	Hours saved × value of time	Faster logistics, shorter flight approaches
Revealed preference	Infer value from behaviour and prices actually paid	Price users pay for correction services
Stated preference (contingent valuation, choice experiments)	Ask people how much they would pay	Willingness to pay for authenticated positioning
Benefit transfer	Reuse values from existing studies, adjusted to the context	Value of a statistical life from national guidance
Value of information	Value of better decisions made possible by better data	Targeted crop treatment, insurance claims, disaster response

Value of information: worked example

A farm cooperative manages 1,000 ha. A disease hits about 20% of fields each season, but nobody knows which ones. Treatment costs €100/ha; an untreated infected field loses €600/ha.

- **Without information**: treat everything (1,000 × €100 = €100,000) or treat nothing (200 ha × €600 = €120,000 losses). Best choice: treat everything, cost **€100,000**.
- **With perfect satellite information**: treat only the 200 infected hectares, cost 200 × €100 = **€20,000**.
- **Value of information** = €100,000 − €20,000 = **€80,000** per season. If the data and analytics cost €5/ha (€5,000), the net value is €75,000.
- With imperfect detection (missed or false alarms) the value falls: a realistic estimate needs the detection accuracy.

Key lesson: **data have value only if they change a decision**. If the farmer would act the same way anyway, the value is zero, whatever the resolution.

3.6 Attribution and additionality

Not every benefit observed after an intervention is caused by it. Three concepts:

- **Deadweight**: what would have happened anyway (in the counterfactual).
- **Displacement**: benefits that are simply moved from one place or company to another.
- **Attribution**: the share of a benefit that is due to this intervention when several factors contribute.

The GNSS attribution problem

Modern receivers combine GPS, Galileo, GLONASS and BeiDou. The benefit of positioning in a segment is **not** the benefit of Galileo. A standard approach:

```
Benefit attributable to Galileo = Total benefit in the segment
  × share of devices that are Galileo-enabled
  × share of the benefit that depends on Galileo specifically
```

Example: precision agriculture generates €500m of annual benefits; 80% of receivers are Galileo-enabled; Galileo's specific contribution (more satellites, HAS, authentication, continuity if GPS degrades) is estimated at 20% → 500 × 0.8 × 0.2 = **€80m**, not €500m.

The last factor is the most uncertain: estimate it with expert interviews, receiver tests and scenarios (e.g. "GPS unavailable for 24 hours"), and test it in the sensitivity analysis.

3.7 Cost-effectiveness and multi-criteria analysis

Cost-effectiveness analysis (CEA)

When benefits cannot be monetised but the objective is fixed, compare options by **cost per unit of result**: e.g. cost per hour of guaranteed timing availability, or the cheapest way to reach a required resilience level.

Multi-criteria analysis (MCA)

Used for strategic choices with several non-monetary dimensions (autonomy, security, time to service). Steps: define criteria; give weights; score each option; compute weighted scores; test how results change with the weights.

Illustrative MCA: options for resilient PNT (scores 1–5, higher is better)

Criterion	Weight A	Weight B	EU-owned LEO-PNT	Buy commercial service	Do-minimum
Resilience	30%	40%	5	4	2
Strategic autonomy	25%	25%	5	2	3
Cost	25%	15%	1	4	5
Time to service	20%	20%	2	4	5
Weighted score, weights A			3.40	3.50	3.60
Weighted score, weights B			3.80	3.50	3.30

Lesson: with weights A the ranking is do-minimum first; with weights B (more weight on resilience, less on cost) the EU-owned option wins. MCA results depend on the weights, so the weights must be justified (e.g. by stakeholder consultation) and their sensitivity shown.

3.8 Economic impact studies

Different from CBA. They answer "how much economic activity does the space sector generate?", not "is it worth it?".

- **Direct** impact: turnover, gross value added (GVA) and jobs of space companies.
- **Indirect**: activity in the supply chain.
- **Induced**: activity from the spending of wages.
- Usually estimated with **input-output multipliers** (Type I = direct + indirect; Type II = + induced).
- Use **GVA**, not turnover, to avoid double counting along the chain.

> *"An impact study tells you how big the activity is; a CBA tells you whether society is better off. I'd keep jobs and industrial turnover out of the NPV and report them separately."*

4. Evaluation and impact assessment

The Commission's **Better Regulation** framework (guidelines and toolbox) sets how EU interventions are designed, assessed and evaluated. Much of the team's public work fits into it.

4.1 The intervention logic



Example for Copernicus uptake actions: *activities* = training and user workshops; *outputs* = number of people trained; *results* = new users actively using Copernicus data; *impacts* = better environmental decisions, new businesses, jobs.

External factors also influence results and impacts: the evaluation must separate the effect of the intervention from them.

4.2 Evaluation (ex post)

Asks whether an existing intervention works. Five **Better Regulation criteria**:

Criterion	Question	Example for Copernicus
Effectiveness	Did it achieve its objectives?	Has the number of non-expert users grown as planned?
Efficiency	At what cost? Were resources used well?	Cost per active user; administrative burden of data access
Relevance	Do the objectives still match the needs?	Do services match what agriculture and insurance need now?
Coherence	Does it fit with other interventions (internal and external)?	Overlap or synergy with national EO programmes and Horizon Europe
EU added value	What does EU action add compared with Member States alone?	Economies of scale, common data policy, single access

> **Trap**
> The OECD-DAC criteria used in development cooperation are relevance, coherence, effectiveness, efficiency, **impact** and **sustainability**. The EU Better Regulation list has **EU added value** instead. Don't mix them.

Evaluation methods

- **Evaluation matrix**: for each evaluation question, judgement criteria, indicators, data sources.
- **Data**: monitoring data, desk research, surveys, interviews, case studies, focus groups, workshops.
- **Triangulation**: answer each question with at least two independent sources.

- **Counterfactual methods** (comparison groups, difference-in-differences) where data allow; otherwise **contribution analysis**: build a plausible chain of evidence that the intervention contributed to the observed change.

4.3 Impact assessment (ex ante)

Prepared before the Commission proposes a significant initiative (e.g. the EU Space Act). Standard structure:

1. **Problem definition**: what is the problem, its drivers, who is affected, how it evolves without action.
2. **Why should the EU act?** Legal basis, **subsidiarity** (can Member States do it alone?) and EU added value.
3. **Objectives**: general, specific, operational; SMART (Specific, Measurable, Achievable, Relevant, Time-bound).
4. **Policy options**, always including the **baseline**.
5. **Impacts**: economic, social, environmental; on SMEs, competitiveness, fundamental rights; administrative burden (the **Standard Cost Model** counts time × frequency × tariff for information obligations). The Commission applies a "one in, one out" approach to new burdens.
6. **Comparison of options**: effectiveness, efficiency, coherence, **proportionality**.
7. **Monitoring and evaluation** arrangements.

Quality control: the **Regulatory Scrutiny Board**. Stakeholders are consulted, typically through a public consultation of 12 weeks.

4.4 KPIs and monitoring frameworks

Essential for the ECA follow-up project. Good KPIs are linked to the intervention logic and have a **baseline**, a **target**, a **source** and a **frequency**.

Level	Weak KPI	Better KPI
Output	Number of events organised	Number of users trained who report using Copernicus data within 6 months
Result	Number of downloads	Number of active users by segment; share of new GNSS devices that are Galileo-enabled; share using HAS or OSNMA
Impact	"Increased awareness"	Revenues of EU downstream companies; estimated socio-economic benefits per segment

> *"I'd build KPIs from the intervention logic, with a baseline and target for each, and favour result indicators like active users and Galileo-enabled market share over activity counts like events or downloads."*

4.5 Gap analysis and user requirements

- **Gap analysis**: current state vs desired state → size of each gap → causes (barriers) → actions to close it → priority.
- **User needs and requirements**: what users need (accuracy, availability, integrity, latency, revisit, price, formats, certification) translated into requirements for the system. EUSPA publishes **User**

Requirements Reports by segment and runs a **User Consultation Platform**.

- Distinguish **needs** (the problem to solve) from **requirements** (measurable performance specifications).

5. Market analysis and forecasting

5.1 Defining the market

Before any number, define: product or service, geography (EU, global), revenue type (devices, services, data, value added), year and currency. Most disagreements between market studies come from different definitions.

- **TAM** (total addressable market): total demand if everyone who could use the service used it.
- **SAM** (serviceable available market): the part you can reach with your offer and channels.
- **SOM** (serviceable obtainable market): the share you can realistically win.

5.2 Sizing approaches

Top-down
Start from a larger known total and apply shares.

```
Market = total market × share relevant to the segment × share of the product
```

Fast, but depends on the quality of the starting figure.

Bottom-up
Build from units, adoption and price.

```
Market = number of potential users × adoption rate × annual spend per user
```

Transparent and testable; requires more data.

Always do both and compare. If they diverge strongly, one of the assumptions is wrong.

Worked example (illustrative assumptions)

Market for lane-level positioning in new cars in the EU in 2034.

- New cars sold per year: assume **10 million** [check the current figure].
- Share with lane-level GNSS (needed for automated driving): **30%** in 2034.
- Extra hardware value per car: **€40**; correction service: **€20** per car per year.
- Hardware market 2034 = 10m × 30% × €40 = **€120m**.
- Service market depends on the **installed base** (cars on the road equipped), not on annual sales: if 15 million equipped cars are active in 2034, services = 15m × €20 = **€300m**. Services end up larger than hardware, consistent with EUSPA's view that services drive GNSS revenues.

5.3 Growth and forecasting

$$\text{CAGR} = (\text{Value}_{end} / \text{Value}_{start})^{1/n} - 1$$

Rule of 72: years to double ≈ 72 / growth rate (%)

Example: €300bn → €580bn in 10 years: $(1.933)^{0.1} - 1 = 6.8\%$. Check with the rule of 72: at 6.8%, doubling takes about 10.6 years, consistent with almost doubling in 10.

Adoption curves and the Bass model

New technologies spread along an **S-curve**: slow start, rapid growth, saturation. The **Bass diffusion model** captures this:

```
New adopters in period t = [ p + q × N(t-1) / m ] × [ m - N(t-1) ]
m = market potential; N = cumulative adopters; p = innovation coefficient (external
influence, e.g. advertising, regulation); q = imitation coefficient (word of mouth)
```

Typical values in the literature: p around 0.01–0.03, q around 0.3–0.5. Regulation (e.g. eCall making GNSS mandatory) acts like a jump in adoption.

Forecasting practice

- **Drivers**: regulation, prices, technology readiness, user awareness, competing solutions.
- **Scenarios**: low, base, high, with explicit assumptions.
- **Calibration**: check the model against historical data and expert views.
- **Consistency**: the installed base must equal past shipments minus retired devices.

5.4 Value chain analysis

Used, for example, in the project on semiconductors for defence and space.

1. Split the chain into stages (for chips: design → fabrication → assembly and test → integration in equipment → platforms).
2. Map the players in each stage (EU and non-EU), their capacity and market share.
3. Identify **dependencies and chokepoints**: single suppliers, non-EU control, export restrictions (e.g. US ITAR), radiation-hardened components.
4. Forecast **demand**: number of platforms (satellites, missiles, aircraft) × components per platform × price, by scenario.
5. Compare supply and demand, and derive policy options.

In defence and space, volumes are small but critical: the issue is **security of supply**, not market size.

5.5 Market and technology monitoring

- Fixed list of **indicators** per segment (shipments, installed base, revenues, prices, penetration of Galileo-enabled devices).
- Recurring **sources**: company reports, industry databases, trade press, patents, start-up funding rounds, standards bodies, regulatory changes.
- **Technology watch**: readiness levels (TRL), new entrants, cost trends.
- **Validation** with experts; documented methodology so updates are consistent.

5.6 Primary research

Surveys

- Define the **population** and the **sampling** method (random, stratified by segment, or purposive when the population is small and specialised).
- Short questionnaire; closed questions for indicators, a few open questions; neutral wording; pilot test.

- Watch for **bias**: self-selection (only enthusiasts answer), social desirability, leading questions.

```
Margin of error (95%, proportion p) = 1.96 × √[ p(1 - p) / n ]
For p = 0.5: n = 385 gives ±5%; n = 200 gives ±6.9%; n = 100 gives ±9.8%
```

In B2B and institutional surveys the population is often small, so results are indicative: combine them with interviews.

Interviews and workshops

- **Semi-structured interviews**: a guide of open questions, same core for everyone, freedom to follow up. Surveys tell you *what*, interviews tell you *why*.
- **Case studies**: in-depth analysis of a few examples to illustrate mechanisms.
- **Workshops and focus groups**: validate findings, prioritise options.
- **Delphi method**: several anonymous rounds of expert estimates with feedback, until views converge. Useful for forecasts with little data.

When sources disagree

1. Check definitions and scope (devices vs services; EU vs global; revenue vs value added).
2. Check year, currency, nominal vs real.
3. Check method and who produced the figure (and their incentives).
4. Reconcile, or present a range, and state clearly which figure you use and why.

6. Market uptake by segment

6.1 Barriers and levers

Barrier	Lever
Low awareness of EU services	Targeted outreach through intermediaries (associations, integrators), demonstrations
Integration cost for manufacturers	Support to chipset and receiver makers; reference designs; funding
No requirement to use it	Inclusion in **regulation and standards** (the strongest lever)
Data hard to use	Ready-to-use products, platforms, APIs, analysis-ready data
Skills gaps	Training, academies, university programmes
Public sector not using it	Public procurement that requires the service; pilots with administrations

Regulation as a lever: examples

- **eCall**: emergency call system mandatory in new car types in the EU since 2018, required to be compatible with Galileo.
- **Smart tachograph** for trucks: uses GNSS, with Galileo and its authentication.
- **MiFID II**: financial market clocks must be synchronised and traceable to UTC, often through GNSS timing.

6.2 Segment snapshots

Segment	Space inputs	Uses and value drivers
Road and automotive	GNSS (HAS, OSNMA), satcom	Navigation, eCall, fleet management, road user charging, connected and automated driving (lane-level accuracy and trusted position). Large volumes; services growing.
Consumer solutions	GNSS (dual frequency)	Smartphones, wearables, location-based services. The largest GNSS segment by devices.
Agriculture	GNSS, EGNOS, HAS; Copernicus (Sentinel-2)	Automatic steering, variable-rate application, crop monitoring, yield forecasting, CAP subsidy checks. Largest EO segment.
Aviation and drones	EGNOS, GNSS, SAR	LPV approaches, U-space for drones; integrity is key.
Maritime	GNSS, Copernicus (Sentinel-1), satcom	Navigation, ship tracking, oil spill detection, fisheries control, search and rescue.
Insurance and finance	EO, GNSS timing	Parametric insurance, claims verification, risk mapping; timestamping of transactions.

Commodities trading	EO, GNSS	Harvest forecasts, oil storage monitoring, port and mine activity, shipping flows: information advantage for traders.
Energy and infrastructure	GNSS timing, EO	Grid synchronisation, monitoring of pipelines and ground motion (InSAR), renewables siting.
Emergency management	Copernicus EMS, Galileo SAR, satcom	Rapid mapping, early warning, communications when networks are down.
Border and internal security	EO, PRS, secure satcom	Surveillance, secure communications; new segment in EUSPA's 2026 report.

> *"The strongest uptake lever is embedding EU services in regulation and standards, as eCall did for Galileo. Communication campaigns alone rarely move adoption."*

7. Excel, data and presentation

7.1 Building a model that survives review

- **Separate sheets**: Inputs (all assumptions, with source and date) → Calculations → Outputs (results, charts).
- **No hard-coded numbers** inside formulas. Every assumption lives in one cell.
- Consistent timeline across columns; one unit per row, clearly labelled (€m, real 2024 prices).
- **Checks**: totals that must match, balance checks, flags for errors.
- Colour convention: inputs in one colour, formulas in another.
- Version control: date and version in the file name; change log.

7.2 Functions you should know

Need	Function / tool
Look up values	XLOOKUP, INDEX + MATCH
Summarise data	Pivot tables, SUMIFS, COUNTIFS, AVERAGEIFS
Discounting	NPV (note: Excel's NPV assumes the first flow is in year 1; add year 0 separately), XNPV, IRR, XIRR, PMT
Growth	RRI (computes CAGR), or (end/start)^(1/n) − 1
Sensitivity	Data tables (one- and two-variable), Scenario Manager, Goal Seek for switching values
Uncertainty	RAND / NORM.INV for a simple Monte Carlo
Data cleaning	Power Query, TRIM, TEXT functions, remove duplicates
Communication	Tornado chart for sensitivity; waterfall chart for benefit breakdown

7.3 Writing and slides

- **Pyramid principle**: start with the answer, then the supporting arguments, then the evidence.
- **Action titles**: each slide title states the conclusion ("GNSS revenues in road will double by 2034, driven by services"), not the topic ("Road market").
- One message per slide; charts labelled with units and sources.
- Executive summary readable on its own; methodology in the annex.
- For non-technical audiences: the result, the three main drivers, what could change it, what is not monetised.

8. AI in consulting work

8.1 Useful applications in this team

- **Desk research**: screening many documents, summarising regulation, finding relevant sources faster.
- **Data extraction**: pulling figures and tables from long PDFs into structured data.
- **Coding**: Python or Excel formulas for data cleaning, analysis and charts.
- **Drafting**: first versions of sections, interview guides, survey questions, slide outlines.
- **Qualitative analysis**: first coding of interview notes and open survey answers into themes.
- **Market monitoring**: scanning news and company announcements to update databases.

8.2 Rules for using it well

- **Verify every fact and number** against a primary source: models can invent figures and references.
- **Confidentiality**: client data and non-public documents go only into approved enterprise tools, never into public ones.
- **Traceability**: keep track of sources so the client can check them.
- **Human judgement**: conclusions and recommendations remain the team's responsibility.
- **Measure the value**: time saved, quality gained. Consultancies increasingly ask juniors to identify, quantify and track the value created by AI.

8.3 EU AI Act in brief

- **Regulation (EU) 2024/1689**, in force since August 2024, with staggered application.
- **Risk-based**: prohibited practices; high-risk systems with strict requirements (data quality, documentation, human oversight); transparency obligations (e.g. disclose AI-generated content); minimal-risk systems largely free.
- Specific obligations for **general-purpose AI models**.
- Prohibitions applied first (from February 2025), then obligations for general-purpose models (from August 2025), then most other rules. **[check]** The Commission has proposed adjustments to the timetable for some high-risk obligations.

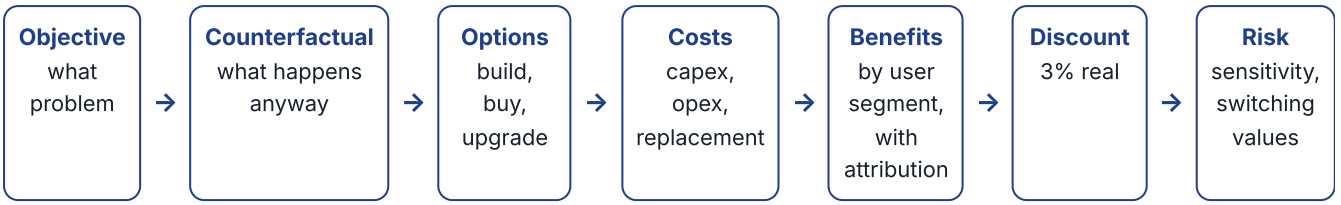
> *"I use AI to go faster on research and first drafts, but every number goes back to a primary source, and client data only go into approved tools. The judgement stays with us."*

9. Interview playbook

9.1 How to answer any technical question

1. **Answer first** in one sentence.
2. Give the **structure**: "I'd look at three things: ...".
3. Go through each point briefly, with one concrete example or number.
4. **Close** with the main risk or caveat: it shows judgement.

9.2 Framework for "how would you do a CBA of X?"



9.3 Framework for "estimate the market for X"

1. Clarify the definition (product, geography, year, revenue type).
2. Choose the driver: users, devices, area (hectares), transactions.
3. Bottom-up: number × adoption × price; state each assumption and round numbers.
4. Sanity check with a top-down figure or a known benchmark.
5. Give a range and the most uncertain assumption.

Mini-case, spoken answer

Q: "How big is the market for satellite-based crop monitoring services for large farms in the EU?"

> *"Let me define it as annual spending by farms above 50 hectares on satellite-based advisory services. Bottom-up: I'd take the number of such farms from Eurostat — say a few hundred thousand — times an adoption rate, perhaps 10 to 15% today, times an annual subscription of a few hundred to a thousand euros. That gives tens of millions of euros a year. I'd cross-check against EUSPA's market report, where agriculture is the largest Earth Observation segment at around a fifth of €3.5 billion. The most uncertain assumption is adoption, so I'd present a range and test it with interviews of service providers."*

9.4 Personal questions: STAR

- **S**ituation: context in one or two sentences.
- **T**ask: your responsibility.
- **A**ction: what *you* did, step by step (the longest part, in the first person).
- **R**esult: outcome with numbers, and what you learned.

Prepare at least: a research task you managed end to end; tight deadline or changing priorities; disagreement; defending your analysis; a mistake; learning a new topic fast; using AI concretely. Write

them in English in the "Personal questions" tab of the trainer.

9.5 Questions to ask them

- What would a typical first project look like for a junior in the team?
- How is time split between desk research, modelling, interviews and client meetings?
- How is the team using AI in market monitoring and reports today?
- Which projects do you expect to grow most in the next two years: secure connectivity, LEO-PNT, the Space Act?
- What distinguishes juniors who do very well in their first year?

9.6 Last-day checklist

- Re-check the **[check]** items: MFF 2028–2034, IRIS² figures, EU Space Act status, AI Act timetable.
- Read the executive summary of the latest EUSPA EU Space Market Report.
- Skim the firm's team page and its recent publications in the sector.
- Rehearse out loud: pitch (90 seconds), CBA structure, market sizing, two STAR stories.
- Prepare one opinion on a recent development (e.g. Celeste, IRIS² acceleration, Space Act).

10. Glossary

Term	Meaning
B/C ratio	Present value of benefits divided by present value of costs
Baseline / counterfactual	Scenario without the intervention
CAGR	Compound annual growth rate
CBA	Cost-benefit analysis
CEA	Cost-effectiveness analysis
CEMS, CAMS, CMEMS, C3S, CLMS	Copernicus services: emergency, atmosphere, marine, climate, land
DG DEFIS	Directorate-General for Defence Industry and Space
EGNOS	European Geostationary Navigation Overlay Service (SBAS)
ENPV / ERR	Economic net present value / economic rate of return
EO	Earth Observation
EUSPA	EU Agency for the Space Programme
FNPV / FRR	Financial net present value / financial rate of return
GEO, MEO, LEO	Geostationary, medium and low Earth orbit
GNSS	Global Navigation Satellite System (GPS, Galileo, GLONASS, BeiDou)
GOVSATCOM	Governmental satellite communications
GVA	Gross value added
HAS	Galileo High Accuracy Service
InSAR	Radar interferometry: measures ground movement
IRIS²	Infrastructure for Resilience, Interconnectivity and Security by Satellite
KPI	Key performance indicator
LPV	Localiser performance with vertical guidance: aircraft approach using SBAS
MCA	Multi-criteria analysis
MFF	Multiannual Financial Framework (EU long-term budget)
NDVI	Normalised difference vegetation index
NPV / IRR	Net present value / internal rate of return
OSNMA	Open Service Navigation Message Authentication
PNT	Positioning, navigation and timing

PPP / RTK	Precise point positioning / real-time kinematic: high-accuracy techniques
PRS	Galileo Public Regulated Service
SAR	Search and rescue; also synthetic aperture radar (context!)
SBAS	Satellite-based augmentation system
SSA / SST	Space situational awareness / space surveillance and tracking
Switching value	Change in a variable that brings NPV to zero
TAM, SAM, SOM	Total, serviceable available, serviceable obtainable market
TRL	Technology readiness level (1–9)
Upstream / downstream	Building space infrastructure / using its signals and data

Sources

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